

```

pythem> set interface
[+] Enter the interface: <wlan-interface>
pythem> set gateway
[+] Enter the gateway: <gateway-id>
pythem> arpspoof start
[+] Setting the packet forwarding.
[+] IPtables redefined.
[+] ARP spoofing initialized.
pythem> sniff
[+] Enter the filter: core

```

Figure 2: Penetration testing using ARP spoofing in PytheM

```

pythem> dhcpspoof start
[+] DHCP Server IP address: <IP-Address>
[+] Broadcast address: <Broadcast Address>
[+] Subnet mask: 255.255.255.0
[+] Router IP address: <Router-IP-Address>
[+] Domain: home
[+] DNS Server IP address: 192.168.1.10 (Fake IP)
[+] DHCP spoofing initialized.
pythem> sniff core

```

Figure 3: Penetration testing using DHCP spoofing and starvation attacks in PytheM

Implementation of ARP spoofing

In an ARP spoofing attack, the malicious attack source sends fake or manipulated ARP (Address Resolution Protocol) messages to the network. This process disguises the router and servers, which allows the attacker to steal information even from a privacy-aware network environment.

Continued from page no...74

The most used option is 'df -iv' which gives information about the free and used blocks.

```

ubuntu@ubuntu:~$ df -iv
Filesystem      Inodes  IUsed  IFree IUse% Mounted on
udev            118658   437   118221    1% /dev
tmpfs           124359    904   123455    1% /run
/dev/sda1       1966080 186081 1779999   10% /
tmpfs           124359     4   124355    1% /dev/shm
tmpfs           124359     6   124353    1% /run/lock
tmpfs           124359    16   124343    1% /sys/fs/cgroup
tmpfs           124359    71   124288    1% /run/user/1000
/dev/sr0         0         0         0    - /media/shashi-
ubuntu/MobileWiFi

```

Other commands

Listed below are commands that are popularly used while writing shell scripts.

1. File operation related commands such as *cat*, *wc*, *touch*, *file*, *which*, *sort*, *cut*, *grep*, *head* & *tail*, *more*, etc
2. IO redirection commands/options (>, <, <<, >>, etc)
3. *who*
4. *tee*
5. Process-oriented commands like *ps*, *nohup*, *kill*, *nice*, *time*, *top*, *batch*, etc

Shell programming

Some samples of small shell scripts using *sh* are listed below.

```

###Sample-1(simple script)###
#!/bin/sh

```

DHCP spoofing attacks or ACK injection

In a DHCP spoofing and starvation attack, the hacker or malicious source can gain access to the DHCP server. By this attack, the attacker can overload the server or important information can be fetched out.

It is always desirable to test the application or network infrastructure before deployment so that there is minimum scope for any cracking or hacking. If the applications or devices are tested using penetration testing and security audit tools like PytheM or any similar library, the overall network environment can be secured from multiple attacks. **END** 

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```

# This is a comment!
echo Hello World      # This is a comment section

###Sample-2(taking decisions)###
#!/bin/bash
if cp $source $target
then
    echo "File copied successfully"
fi

###Sample-3(simple for loop)###
#!/bin/bash
for number in 1 2 3 4 5
do
    echo $number
done
exit 0
# this prints the numbers

```

Each shell will support a 'shell prompt' and we can set this prompt using the variable 'PS1'.

For example, you can execute this command on any shell prompt `$PS1="What_prompt#"`.

Now the prompt will be set to 'What_prompt#'.

Watch out for more on shell scripting in future issues of OSFY. **END** 

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